LQC Whitepaper

Official Review Edition

Modular Global Liquidity Infrastructure

Connecting Fragmented Web3 Liquidity

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Unaudited testnet MVP · No production or listing claim

Contents and Status Legend

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Status legend: Implemented means public repository code/test evidence. In development means active implementation or configuration work. Subject to review means independent, legal, governance or deployment verification is required. Planned means no completed production module is claimed.

1. Executive Summary

Liquidity Chain (LQC) is a modular infrastructure project for connecting fragmented liquidity across decentralized exchanges and, in later phases, across blockchain ecosystems. The initial engineering focus is an EVM routing stack and LQC Flow automated market maker for controlled BNB Smart Chain testnet validation.

The public repository includes protocol-neutral adapter registration, failure-isolated quotes, gas-aware route comparison, one-to-four-route split optimization, atomic protected execution, native BNB handling, risk limits, emergency controls and testnet deployment validation.

Current status: unaudited testnet MVP. No production deployment, live liquidity, exchange listing, audit approval or guaranteed commercial outcome is claimed.

- Liquidity first - utility second - revenue third - scale last.
- Security gates precede unrestricted deployment.
- Every material claim should be independently verifiable.

2. The Liquidity Fragmentation Problem

Liquidity is distributed across exchanges, pools, fee tiers, networks and assets. A visible quoted price may not represent the best executable result once depth, price impact, pool fees, gas and failure risk are considered.

Users and applications often perform fragmented discovery and submit an order to one venue. This can produce unnecessary slippage, failed execution and inconsistent operational controls.

Fragmentation	Resulting challenge
Many venues	Quotes and interfaces differ
Uneven depth	Price impact varies by order size
Different networks	Assets and messages require separate security assumptions
Operational risk	Paused or compromised routes must be isolated

3. LQC's Proposed Response

LQC separates route discovery, optimization, execution and risk control into auditable modules. Reviewed adapters translate protocol-specific behavior into a shared interface while the registry determines which integrations are eligible.

The objective is to compare executable outcomes, support direct and multi-hop paths, split larger orders when justified, and apply protection before funds are sent to an external venue.

- Multi-DEX liquidity aggregation
- Gas- and slippage-aware route selection
- Atomic single or split execution
- Staged limits and emergency disable controls
- Public deployment and verification evidence

4. Purpose, Users and Differentiation

LQC is designed as liquidity infrastructure rather than a single isolated exchange. Its purpose is to connect fragmented DEX liquidity, compare executable outcomes and provide a controlled foundation for liquidity deployment and later DeFi services.

Stakeholder	Intended value
Traders and wallets	Better route discovery with gas, slippage and price-impact awareness
Liquidity providers	Infrastructure for measured allocation across approved pools
DEX ecosystems	Adapter-based access without replacing the shared routing stack
Projects and market makers	Transparent execution controls and auditable route policy

LQC distinction	Reviewer evidence
Protocol-neutral adapters	Separate V2, V3 and LQC Flow integration modules
Risk before execution	Token, transaction, daily and DEX-specific caps
Atomic route protection	Deadline, minimum output, exact approval and rollback tests
Evidence-led launch	Pinned tests, deployment records, validation and verification tooling

The commercial objective is sustainable infrastructure usage. The whitepaper does not rely on token-price appreciation or guaranteed exchange listing as a value proposition.

5. Principles and Modular Architecture

Principle	Engineering application
Liquidity first	Executable depth and measurable routing quality
Modularity	Separate quote, execution, risk and integration responsibilities
Least privilege	Divide governance, emergency and risk authority
Evidence	Pin code, settings, addresses and on-chain state
Phased launch	Testnet, caps and independent review before expansion

Layer	Current status
LQC Flow AMM	Implemented testnet MVP
Registry / quote / optimizer	Implemented
Execution / native BNB	Implemented
Risk and emergency controls	Implemented foundation
Lending / bridge / perpetuals	Planned as separate scope

Production configuration, audits, external integrations and live liquidity are separate gates from repository implementation.

6. LQC Flow AMM

LQC Flow is the native constant-product AMM implementation in the repository. Its factory creates token pairs, each pair holds reserves and issues ERC-20 liquidity shares, and the router coordinates liquidity and swaps.

- Constant-product reserve model with a 0.30% swap fee retained for LPs
- Minimum liquidity permanently locked
- Exact-input and exact-output swaps
- Direct and multi-hop paths
- Token/native-BNB liquidity and swap support
- Deadline and minimum-output protection

LQC Flow is testnet-oriented and unaudited. It must not be used with production funds before independent review and formal launch approval.

7. Adapter and Registry Model

Each reviewed DEX integration uses a protocol-specific adapter. The registry records adapter identity, priority and active state. A failed quote is isolated so another enabled route may still be evaluated.

Protocol-neutral architecture does not mean automatic compatibility. Every integration requires address review, path validation, automated tests and security review before activation.

Integration	Implemented scope
LQC Flow	Native direct/multi-hop quote and execution
PancakeSwap V2	Compatible-router quote and protected exact-input execution
PancakeSwap V3	Packed paths, reviewed fee tiers/pools and maximum three hops

8. Quote Discovery and Gas-Aware Selection

The quote layer queries enabled adapters independently and excludes failed calls from comparison. Deterministic registry priority resolves ties and makes selection reproducible.

Expected output is only one component of execution quality. The intended comparison considers pool fee, price impact, route gas, minimum output, deadline and route availability.

A quote is an estimate, not a guarantee. Realized output depends on state changes, inclusion timing, network conditions and user-selected protection.

The gas-cost oracle converts estimated BNB gas into units of the output token. The router can therefore compare estimated net output rather than gross quote alone.

- Primary and secondary feed checks
- Positive-value enforcement
- Freshness windows
- Maximum feed-deviation rule
- Governance-controlled configuration

Production oracle addresses, heartbeat windows, deviation thresholds and fallback policy are not finalized. The current implementation uses test infrastructure and does not guarantee best execution.

9. Split Optimization

The optimizer samples allocations and may distribute one order across two to four eligible routes when the estimated net result is superior. Allocation and expected-output totals must reconcile exactly.

The auto router converts the optimizer result into an atomic execution plan and derives route-level minimum outputs from the user's slippage tolerance.

- One-to-four active route cap
- No duplicate DEX identifiers
- Deterministic allocation and tie behavior
- Inactive legs excluded
- Invalid part and route bounds rejected

10. Protected Execution and Native BNB

The execution router transfers the specified input, grants an exact temporary approval to the selected adapter and verifies recipient output. Deadline and minimum-output conditions apply to supported exact-input flows.

For split execution, every leg is part of one transaction. If a later adapter fails, earlier transfers, pool changes and risk usage revert.

Repository tests require zero residual router/adapter custody after successful and reverted paths. This tested property is not a substitute for an independent audit.

The native router wraps BNB into WBNB before protected token execution and unwraps WBNB for the final native-asset transfer. It is designed not to retain user balances.

- Configured WBNB address
- Controlled receive path
- Minimum-output and deadline enforcement
- Exact approval lifecycle
- No unsolicited native transfer acceptance

Each deployment must verify the canonical WBNB address and module linkage on the selected network.

11. Risk, Governance and Emergency Controls

The risk registry stages exposure through token allowlisting and input caps. Limits apply before adapter execution and rejected transactions must not consume daily capacity.

Control	Purpose
Token allowlist	Restrict eligible input assets
Per-transaction cap	Bound single-order exposure
Per-DEX/token cap	Limit venue-specific exposure
UTC-day cap	Bound aggregate daily input
Module pause	Stop new protected execution

Structural expansion is reserved for timelocked governance. Guardians may disable a DEX or pause new swaps immediately, but they cannot resume service, move user funds, add adapters or increase limits.

A designated risk role may reduce existing limits but cannot create a new permission or expand exposure. Emergency-controller ownership uses nomination and explicit acceptance.

Production governance configuration is in development. Final multisig addresses, signer threshold, timelock delay and operating procedures require formal approval and verifiable disclosure.

12. Token Compatibility Boundaries

The current execution design supports standard ERC-20 accounting assumptions. Fee-on-transfer inputs are explicitly rejected when the received balance differs from the requested input.

- Fee-on-transfer execution: unsupported and rejected
- Rebasing token execution: unsupported
- Permit signatures: planned for later review
- Non-standard approval behavior: integration review required
- Token and pool allowlists: subject to production approval

Unsupported behavior must not leave retained balances, approvals or risk-accounting residue.

13. Reproducible Engineering Baseline

At the publication baseline, the repository compiles 35 Solidity source files and reports 53 passing automated tests using locked dependencies. The exact Git commit and its CI run are authoritative when the suite evolves.

Area	Representative evidence
AMM	Reserves, product, LP lock, liquidity and swap flows
Router	Quotes, adapters, gas, split and rollback
Risk	Allowlists, caps, model-based daily accounting
Governance	Timelock, pause-only guardian, ownership transfer
Deployment	Chain 97, bytecode, ownership and module links

Passing tests do not prove the absence of vulnerabilities or certify production economic safety.

14. BSC Testnet Deployment

Deployment tooling targets BSC testnet chain ID 97. It deploys test tokens, LQC Flow and Router 2.0 modules, establishes ownership and registers the native adapter. Reviewed PancakeSwap endpoints may be supplied separately.

- Read-only preflight checks chain ID, contract bytecode, reviewed endpoints and deployer tBNB balance
- Reject unexpected chain IDs before deployment
- Record deployed contracts and external dependencies
- Record compiler and optimizer settings
- Record governance and adapter configuration
- Never commit deployer private keys

A generated deployment record is evidence only when its addresses and transactions are independently checked on the explorer.

15. Address and Explorer Verification

The read-only validator checks deployed bytecode, Router-to-Factory and WBNB relationships, registry order, adapter status, module linkage, ownership and minimum timelock delay.

The verification-package generator produces Solidity Standard JSON input, compiler settings, constructor arguments, deployed addresses and the recorded source revision for BscScan submission.

Explorer verification is the next deployment evidence gate. It becomes complete only after a real deployment record is validated and published against the exact reviewed commit.

16. Lending, Oracle and Liquidation Direction

Lending is a future module and is not part of the current production claim. The design direction covers approved collateral deposits, borrowing, repayment, health-factor monitoring and controlled liquidation.

Parameter	Current design disclosure
Maximum LTV	45% baseline
Liquidation threshold	60%
Base liquidation penalty	5%
Automatic additional borrowing	Off by default; explicit opt-in required

Final values require economic simulation, oracle validation, liquidity testing, audit, legal review and formal governance approval.

Account risk should be determined by collateral value, debt and health factor rather than price movement alone. A shallow LQC pool must not be the sole lending oracle.

- Validated external feeds
- Sufficiently liquid DEX TWAP where appropriate
- Freshness and deviation checks
- Debt and market-specific caps
- Partial liquidation where safe and economical
- Emergency pause and recovery procedures

Production liquidation requires adversarial simulation, keeper design, bad-debt handling and clearly disclosed parameters.

17. Cross-Chain Direction

Cross-chain expansion is phased future scope. Non-EVM venues are not assumed compatible with EVM adapters and must use a separately reviewed messaging and accounting layer.

- Authenticated messages
- Nonce consumption and replay protection
- Rate limits
- Emergency pause
- Canonical asset identification
- Global supply reconciliation

Cross-chain movement must not create unexplained LQC supply. Bridge design and audit are separate from Router 2.0 review.

18. Utility, Fees, Treasury and Burn

The approved design direction associates LQC with future loan-origination and repayment fee settlement and permanent burning of those fees. This does not mean the lending or burn contracts are currently live.

Routing and AMM development does not by itself activate staking, governance, fee benefits, gas support, treasury distributions or other proposed utilities.

Utility activates only after contract completion, audit, parameters, legal review, formal approval and on-chain disclosure.

Potential revenue sources include routing and swap fees, future lending and repayment fees, liquidation fees, cross-chain services and Liquidity-as-a-Service arrangements. Actual revenue depends on deployed products and usage.

Fee rates, settlement, conversion, burn addresses, treasury allocation and distributions are not final. A future burn must be verifiable on-chain and permanently remove the relevant LQC from circulation.

No yield, revenue, token-price appreciation or investment return is promised.

19. Tokenomics and Release Framework

The current approved project design uses a fixed maximum supply of 1,000,000,000 LQC and planned TGE circulation of 150,000,000 LQC (15%). Earlier draft figures are superseded.

Allocation	Share	LQC
Future ecosystem rewards	35%	350,000,000
Community initial	20%	200,000,000
Team and core contributors	20%	200,000,000
Protocol treasury	10%	100,000,000
Liquidity and market making	10%	100,000,000
Grants and strategic ecosystem	5%	50,000,000

Allocation	TGE / release framework
Ecosystem rewards	0 at TGE; 7+ years, up to 50M annually
Community initial	80M at TGE; remaining 120M activity-based over 24 months
Team / contributors	0 at TGE; 12-month cliff, then 36-month monthly vesting
Protocol treasury	10M at TGE; remaining 90M under 5-year budget framework
Liquidity / MM	50M at TGE; remaining 50M linked to exchange and pool growth
Grants / strategic	10M at TGE; remaining 40M milestone-based

The planned 150M TGE circulation is Community 80M, Liquidity/MM 50M, Treasury 10M and Grants 10M. Contract and wallet evidence will be published after implementation and verification.

20. Supply and Vesting Verification

Before launch, the token contract cap, total supply, decimals and mint/burn powers must reconcile to the approved disclosure. Every allocation wallet and vesting contract should be labeled and independently verifiable.

- Canonical network and token address
- Deployment and ownership transactions
- Explorer-verified source and ABI
- Mint, pause, blacklist, upgrade and recovery powers
- Beneficiaries, cliffs and vesting schedules
- Circulating-supply methodology and excluded wallets

No on-chain production token evidence is claimed in this edition.

21. Development Status and Delivery Plan

Workstream	Current status / next gate
LQC Flow AMM	Implemented testnet MVP
Router 2.0 and adapters	Implemented; production integration review required
Risk / governance	Implemented foundation; finalize production roles
BSC testnet tooling	Implemented; publish real deployment evidence
Independent review	Audit, remediation and retest required
Future modules	Separate design, implementation, testing and audits

Phase	Gate
Controlled testnet	Addresses, verified source, smoke tests and capped routes
Independent review	Audit, remediation, retest and known-issues disclosure
Capped pilot	Multisig, oracle, monitoring, legal and liquidity approval
Expansion	Measured reliability, depth and incident-free operation

22. Audit and Incident Readiness

An independent audit engagement must pin the exact commit, compiler, settings, contracts, addresses and configuration reviewed. Every Critical and High finding requires documented remediation and auditor retest or an explicit unresolved disposition.

- Named incident responders and escalation contacts
- Alert coverage and pause procedure
- Compromised-key and signer-loss procedure
- Public vulnerability reporting policy
- Postmortem and recovery policy
- Periodic emergency drills

The repository's audit scope and handoff documents are preparation materials, not an audit report.

23. CEX Listing Evidence

Evidence group	Required before submission
Technical	Pinned build, tests, audit, verified source and addresses
Token	Supply, decimals, powers, vesting and circulation proof
Market	Timestamped liquidity, holders, volume and methodology
Operations	Deposits, withdrawals, confirmations, monitoring and contacts
Legal	Entity, KYB/KYC, legal analysis, sanctions and AML controls
Integrity	Market-making policy and wash-trading prohibition

Evidence not yet available must be completed, marked not applicable with accepted reasoning, or disclosed as an unresolved risk. No exchange approval is claimed.

24. Team and Corporate Accountability

LQC is organized through MMXlabs&LQC; LLC, a Wyoming limited liability company filed on April 30, 2026. The public website identifies the project's leadership, marketing and advisory structure; exchange KYB and beneficial-ownership evidence must be delivered through the reviewer's secure process.

Role	Publicly disclosed member
Co-Founder	ALI ISIK
Co-Founder	CHEONHO KIM
Chief Marketing Officer	AHMET DIZLEK
Advisor	JEONG JAE-WON
Engineering	Four-member technical function disclosed on the project website

- Corporate authority and authorized signatory must match the exchange application
- Engineering, security, treasury and market-operation responsibilities must have named owners before submission
- Team identity evidence should be shared securely and reconciled with public profiles
- No private personal documents should be committed to the public repository

Public team pages support transparency but do not replace exchange KYC/KYB, background review or signed authority records.

25. Legal, Compliance and Immediate Priorities

The disclosed project entity is MMXlabs&LQC; LLC, a Wyoming limited liability company. Formation, good standing, managers, beneficial ownership, authorized signatory, KYB and legal-opinion materials must be supplied through the reviewer's secure process.

LQC has progressed from architecture into a testable EVM routing stack with a native AMM, external DEX adapters, gas-aware comparison, atomic split execution and staged risk controls.

Priority	Next evidence milestone
1	Controlled BSC testnet deployment and explorer-verified source
2	Final multisig, roles, oracle, allowlists and operational limits
3	Independent audit, remediation and retest
4	Token, vesting, circulation, legal and CEX evidence packages
5	Monitored capped-liquidity pilot before expansion

This document is technical and informational. It is not investment, legal, financial or tax advice; an offer or solicitation; or a guarantee of launch, listing, liquidity, revenue or return.

Digital assets and DeFi involve smart-contract, oracle, bridge, liquidity, market, operational, cybersecurity and regulatory risks.